pharmaceuticals, plastics, inks, batteries, textiles and electrical equipment. Zinc sulphide is used in making luminous paints, fluorescent lights and x-ray screens.

- Brass an alloy of zinc is used in water valves and communication equipment as it is highly resistant to corrosion.

1.7.3 Applications of Fe

- Iron is one of the most useful metals and its alloys are used everywhere including bridges, electricity pylons, bicycle chains, cutting tools and rifle barrels.
- Cast iron is used to make pipes, valves and pumps stoves etc...
- Magnets can be made from iron and its alloys and compounds.
- An important alloy of iron is stainless steel, and it is very resistant to corrosion. It is used in architecture, bearings, cutlery, surgical instruments and jewellery. Nickel steel is used for making cables, automobiles and aeroplane parts. Chrome steels are used for maufacturing cutting tools and crushing machines

1.7.4 Applications of Cu

Copper is the first metal used by the human and extended use of its alloy bronze resulted in a new era,'Bronze age'

Copper is used for making coins and ornaments along with gold and other metals.

Copper and its alloys are used for making wires, water pipes and other electrical parts

1.7.5 Applications of Au

- Gold, one of the expensive and precious metals. It is used for coinage, and has been used as standard for monetary systems in some countries.
- It is used extensively in jewellery in its alloy form with copper. It is also used in electroplating to cover other metals with a thin layer of gold which are used in watches, artificial limb joints, cheap jewellery, dental fillings and electrical connectors.
- Gold nanoparticles are also used for increasing the efficiency of solar cells and also used as catalysts.

XII U1 Metallurgy - Jerald.indd 18 16-12-2022 18:53:03